A SEPARABLE ALGEBRAIC RESIDUE FIELD EXTENSION IS REALISED BY A COLIMIT OF FINITE ETALE EXTENSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The extension constructed for a prescribed residue field. Used at the start of the counted proof (lines 44869-44874). Provides a well-ordered colimit presentation whose successor maps are finite etale.
- Transfinite induction over the well-ordered index set. Used throughout the counted proof (lines 44874-44888). Reduces the statement to successor and limit steps, each handled by a separate argument.

Source reference. Original lemma 03C3: sources/stacks-algebra/pinned-original-source.tex, source lines 44790–44790. Navigation pointer for \ref{lemma-flat-local-given-residue-field} used inside the counted span (line 44870); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $(R, \mathfrak{m}, k)$ be a local ring. If $k \subset K$ is a separable algebraic extension, then there exists a directed set I and a system of finite étale extensions $R \subset R_i$, $i \in I$ of local rings such that $R' = \operatorname{colim} R_i$ has residue field K (as extension of k).*

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Proof. Let $R \subset R'$ be the extension constructed in the proof of Lemma 03C3. By construction $R' = \operatorname{colim}_{\alpha \in A} R_\alpha$ where A is a well-ordered set and the transition maps $R_\alpha \to R_{\alpha+1}$ are finite étale and $R_\alpha = \operatorname{colim}_{\beta<\alpha} R_\beta$ if α is not a successor. We will prove the result by transfinite induction.

Suppose the result holds for R_α, i.e., $R_\alpha = \operatorname{colim} R_i$ with R_i finite étale over R. Since $R_\alpha \to R_{\alpha+1}$ is finite étale there exists an i and a finite étale extension $R_i \to R_{i,1}$ such that $R_{\alpha+1} = R_\alpha \otimes_{R_i} R_{i,1}$. Thus $R_{\alpha+1} = \operatorname{colim}_{i' \geq i} R_{i'} \otimes_{R_i} R_{i,1}$ and the result holds for $\alpha + 1$. Suppose α is not a successor and the result holds for R_β for all $\beta < \alpha$. Since every finite subset $E \subset R_\alpha$ is contained in R_β for some $\beta < \alpha$ and we see that E is contained in a finite étale subextension by assumption. Thus the result holds for R_α. $\square$